

Machine Learning

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About me

- PhD in Machine Learning in 2020
- Lecturer at CentraleSupélec/Paris Saclay for 7 years now
- I own a (very small) AI consulting company
 - ▶ Some of my current or former clients:

SAMSUNG



L'ORÉAL



LVMH



- Also co-founded 2 startups

Contact

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- Personal email address: yannicklecacheux@gmail.com
 - ▶ For urgent matters
 - ▶ Or to keep in touch after the course

About the class

- 8 sessions, ~ 3 hours each
 - ▶ divided into lecture - break - lab

Lab sessions

Please bring your laptop for the lab session

Course evaluation (subject to change):

- $\sim 25\%$ of the final grade is the labs
 - ▶ Labs are due one week after they are issued (so Friday at $\sim$ midnight)
- $\sim 75\%$ is the final project
 - ▶ Grade for final project is $\sim 50\%$ score, $\sim 50\%$ report

Project

More info about project coming soon

A word about AI

AI rules

As a general rule, the use of AI is **not permitted** in this course.

Official policy:

- Evidence of AI use on any part of an assignment warrants a grade of 0 for the entire assignment.
- A repeated offense warrants a failing grade for the entire course.

Additional policy

- The teacher may sometimes opt for a more lenient approach at their sole discretion, with no obligation whatsoever.
- Accusations of AI use may be challenged by students who believe they were wrongly accused. Doing so will trigger a re-evaluation with a strict application of the official policy.

Taking this class means you agree to these rules.

If you have doubts about whether these rules apply to a specific situation (e.g. to help plot a complicated figure), please ask. By default, the rules above apply to everything.

Course prerequisites

I am assuming you are at least somehow familiar with:

- **Math:**

- ▶ probability & statistics (bayes theorem, maximum likelihood...)
- ▶ linear algebra (matrix multiplication, eigenvalues, eigenvectors...)
- ▶ calculus (derivatives, gradients, gradient descent...)

- **Programming:** ideally Python

I can recommend readings if necessary.

We will briefly cover some (potentially) new material such as:

- Matrix calculus
- Constrained optimization (i.e. Lagrange multipliers)

Refresher material

Recommended: doing exercises in 1st part of (optional) homework 1 as a refresher on e.g. maximum likelihood and practice for new concepts e.g. matrix calculus.

Optional work and readings

Further readings:

- *Pattern Recognition and Machine Learning*, by Christopher Bishop
- *Mathematics for Machine Learning*, by Marc Peter Deisenroth et al.
- *Deep Learning*, by Ian Goodfellow et al.

Optional practice exercises:

- Homeworks 1 and 2
- Other useful exercises: <https://www.deep-ml.com/>

Additional material

Some useful resources and exercises may be found on my blog (WIP):
<https://manifold.fr/blog>

Set up for labs

Setting up a Python Environment with Conda:

- 1 Download and install Miniconda:
`https://docs.anaconda.com/miniconda/`
- 2 Create a new Conda environment:
`conda create -n mlclass python=3.11`
- 3 Activate the environment:
`conda activate mlclass`
- 4 Install necessary packages using pip:
`pip install scikit-learn jupyter matplotlib`
- 5 Launch Jupyter Notebook:
`jupyter notebook`

Alternatively, you may use Google Collab, Visual Studio Code, or any other method that lets you read, modify and run iPython Notebooks

Other

Course materials

Pending full access to Edunao, course materials will be available here:
<https://github.com/yannick-lc/mlclass2026>

- **Please do not share the slides publicly** (I don't have the rights to all the images I used in the slides)
- Also, please also don't share the solutions to labs and homework assignments
- Most content is on slides, but not all. I will sometimes explain things orally/on the board. You may want to take notes.
- If at any point something is not clear or you want further information, **please do ask questions.**